

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCOC701 - ARTIFICIAL INTELLIGENCE

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 What is the primary goal of Artificial Intelligence (AI)?
 - A) To create software that operates without user input
 - B) To develop machines that think and act rationally
 - C) To replace all human jobs with robots
 - D) To simulate physical environments in virtual reality
 - 2 Which of the following best describes an intelligent agent?
 - A) A system that solves all problems independently
 - B) A system that perceives its environment and takes actions to achieve goals
 - C) A system that operates only in controlled environment
 - D) static program designed for a specific task
 - 3 Which type of environment is the easiest for an AI agent to operate in?
 - A) Fully observable and deterministic
 - B) Partially observable and stochastic
 - C) Dynamic and multi-agent
 - D) Partially observable continuous
 - 4 Which search strategy expands nodes level by level without considering path cost?
 - A) Depth-first search
 - B) Uniform- cost search
 - C) Breadth- first search
 - D) Greedy best- first search
 - 5 What is state space?
 - A) The whole problem
 - B) Your Definition to a problem
 - C) Problem you design
 - D) Representing your problem with variable and parameter

- 6 Which of the following is an example of a heuristic function in AI?
 - A) The shortest path in a graph
 - B) The Manhattan distance in a grid problem
 - C) A binary search algorithm
 - D) A depth-first traversal
- 7 Knowledge and reasoning also play a crucial role in dealing with ----- environment.
 - A) Completely Observable
 - B) Partially Observable
 - C) Neither Completely nor Partially Observable
 - D) Only Completely and Partially Observable
- 8 Which reasoning approach works backward from the goal to determine the necessary steps?
 - A) Forward reasoning
 - B) Backward reasoning
 - C) Deductive reasoning
 - D) Inductive reasoning
- 9 What is the primary purpose of Bayesian networks in AI?
 - A) To represent deterministic relationships between variables
 - B) To handle reasoning under uncertainty
 - C) To perform hierarchical planning
 - D) To implement goal stack planning
- 10 Semantic analysis in NLP focuses on:
 - A) Sentence boundary detection
 - B) Parsing sentence structure
 - C) Extracting meaning from text
 - D) Managing discourse relations
- 11 Which is not Familiar Connectives in First Order Logic
 - A) and
 - B) iff
 - C) or
 - D) not
- 12 Where does the bayes rule can be used?
 - A) Solving queries
 - B) Increasing complexity
 - C) Decreasing complexity
 - D) Answering probabilistic query

Q2. Solve the following.

- A) Explain the following in detail. i) Goal Based Agents ii) Utility Based Agents. 6
- B) Elaborate Artificial intelligence with suitable example along with it's Application. 6

Q3. Solve the following.

- A) What is alpha-beta pruning? Explain the function of alpha-beta pruning 6
- B) Give the outline of Uniform cost search algorithm. 6

Q4. Solve Any Two of the following.

- A) What is Quantifier? Differentiate in between universal quantifier & Existential quantifier with its example. 6

- B) Give outline of simple forward chaining algorithm with suitable example. 6
 - C) List and Explain the issues in knowledge representation. 6
- Q5. Solve Any Two of the following.
- A) What are different components of planning system. 6
 - B) Explain about Dempster- shafer theory. 6
 - C) List out the planning terminologies and components of planning. 6
- Q6. Solve Any Two of the following.
- A) Explain Semantic analysis in NLP. 6
 - B) What is Expert system? Explain its Architecture, Features & applications. 6
 - C) Comparison between Syntactic Processing & Semantic Processing. 6

***** End *****